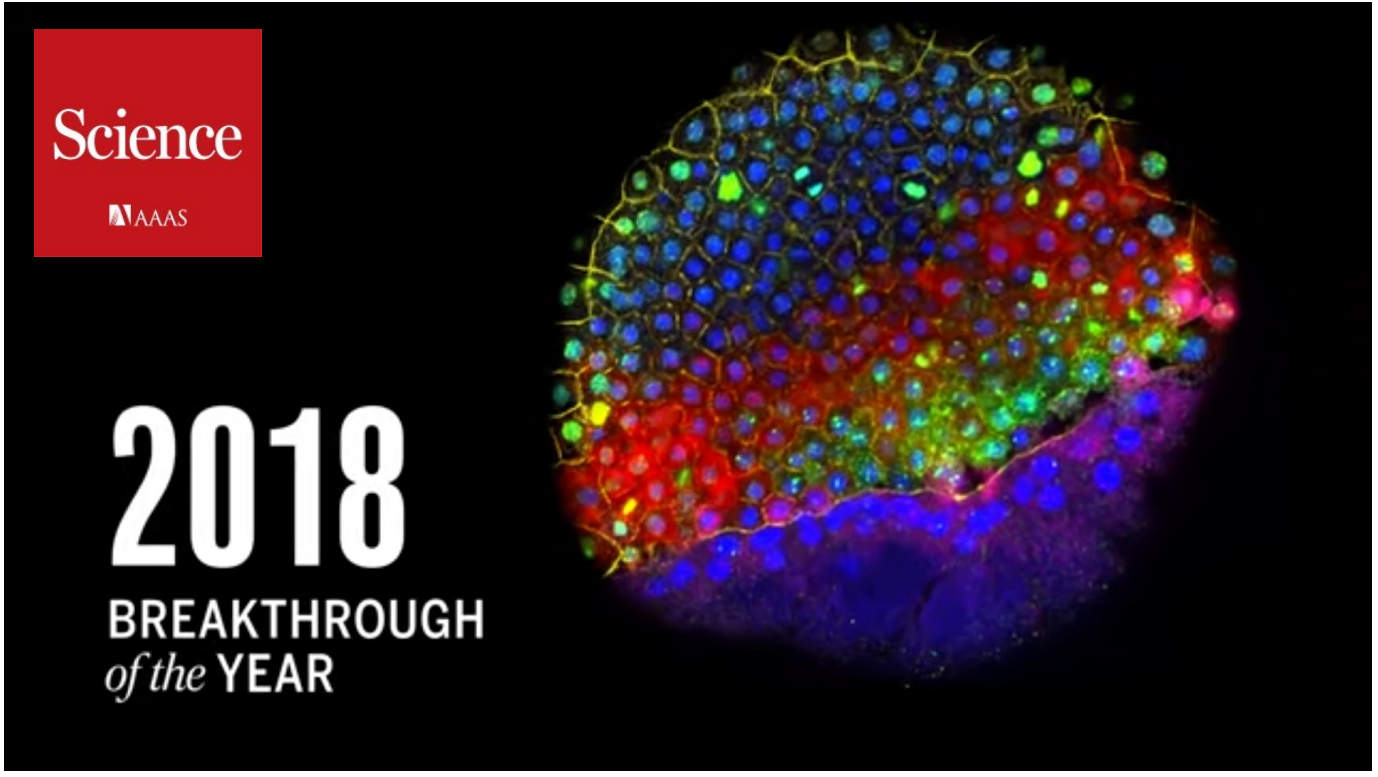
Single Cell RNAseq Analysis

Nikolay Oskolkov, Metabolic Research Group leader at NIRI, Riga, Latvia

PR Stats course, **Single-Cell RNA Sequencing course outline**



@NikolayOskolkov

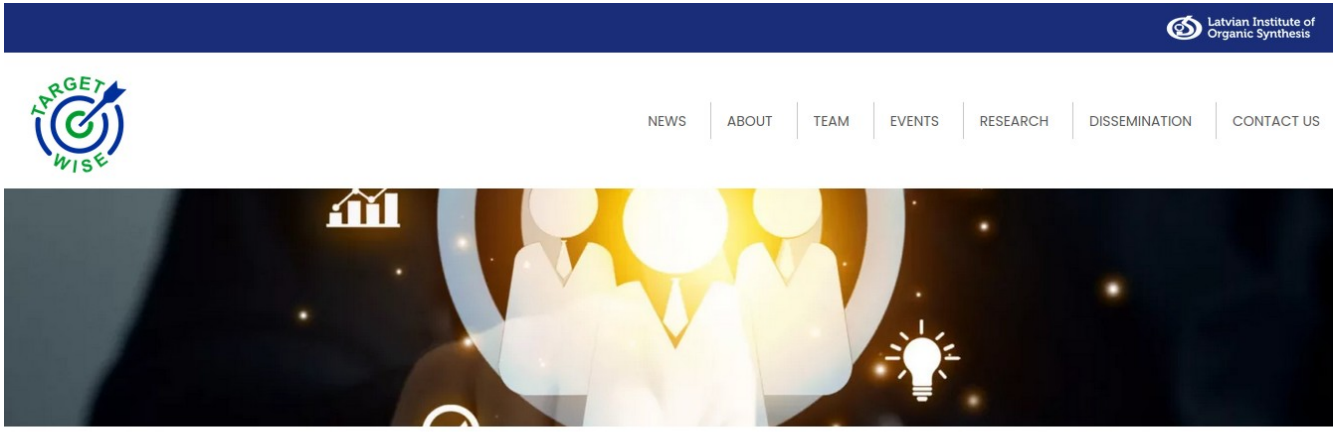
@oskolkov.bsky.social

Personal homepage:
https://nikolay-oskolkov.com

Nature method of the year:
2019 (single cell multi-omics),
2020 (spatial transcriptomics)

Image adapted from Science, 362, 6421, 2018

Metabolic Research Group at NIRI



1 more postdoctoral fellow and 1 PhD student to be hired

If you know anyone who might be interested, please contact me!



PhD. Nikolay Oskolkov
Group Leader (PI) of the Metabolic Research Group



Mohammad Banabazi, postdoc in Metabolic Research Group



Kristina Grausa, PhD student in Metabolic Research Group



Daniel Rivas, MD, postdoc in Metabolic Research Group

About us

Organizer: Oliver Hooker, PR Stats courses

oliver@prstats.org

Instructor:

Dr. Nikolay Oskolkov, MRG Group Leader, NIRI, Riga,
lecturer at Lund University, Lund,Sweden

nikolay.oskolkov@niri.lv



@NikolayOskolkov

@oskolkov.bsky.social

Personal homepage:
https://nikolay-oskolkov.com

Brief introduction: who am I

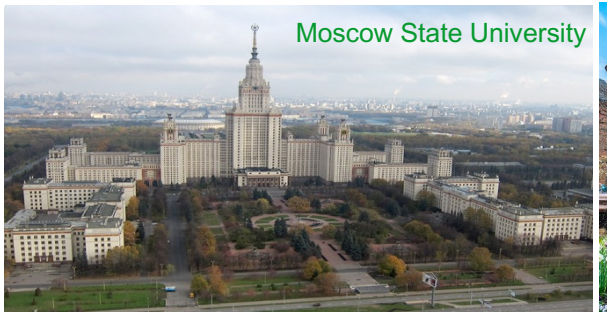
2007 PhD in theoretical physics in Moscow, Russia

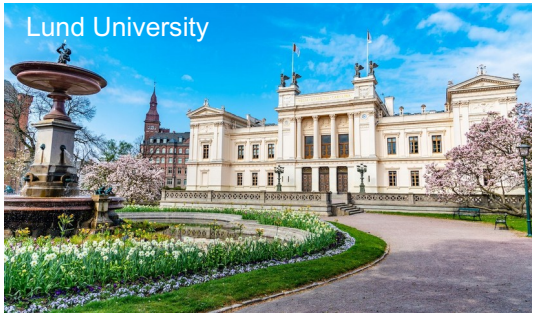
2011 medical genomics at Lund University, Sweden

2016 bioinformatician at NBIS SciLifeLab, Sweden

2025 Metabolic Research Group leader, NIRI, Latvia

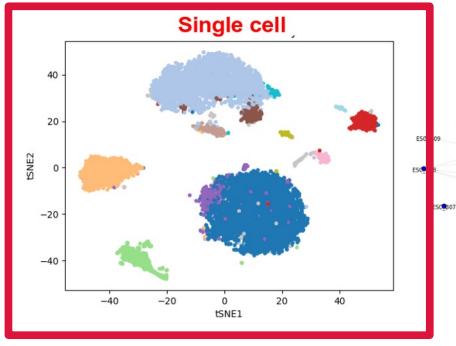
60+ publications; h-index = 28; 4,600+ citations

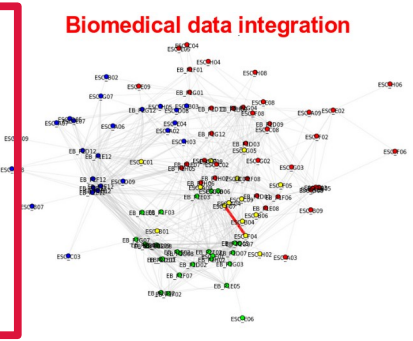


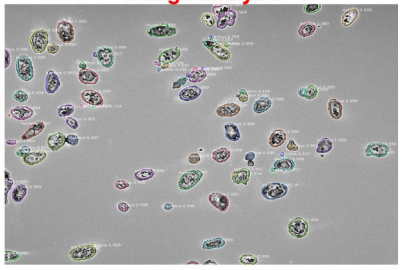


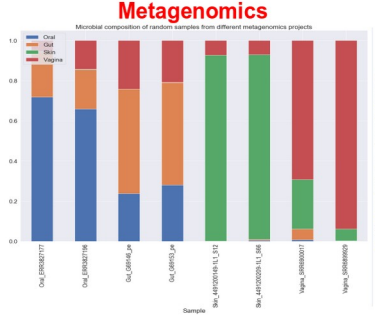


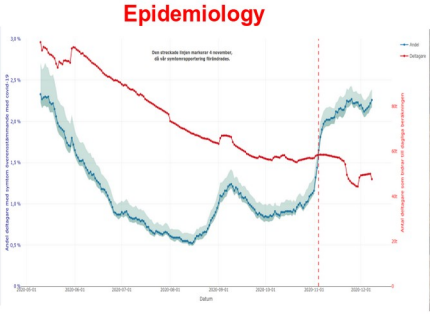












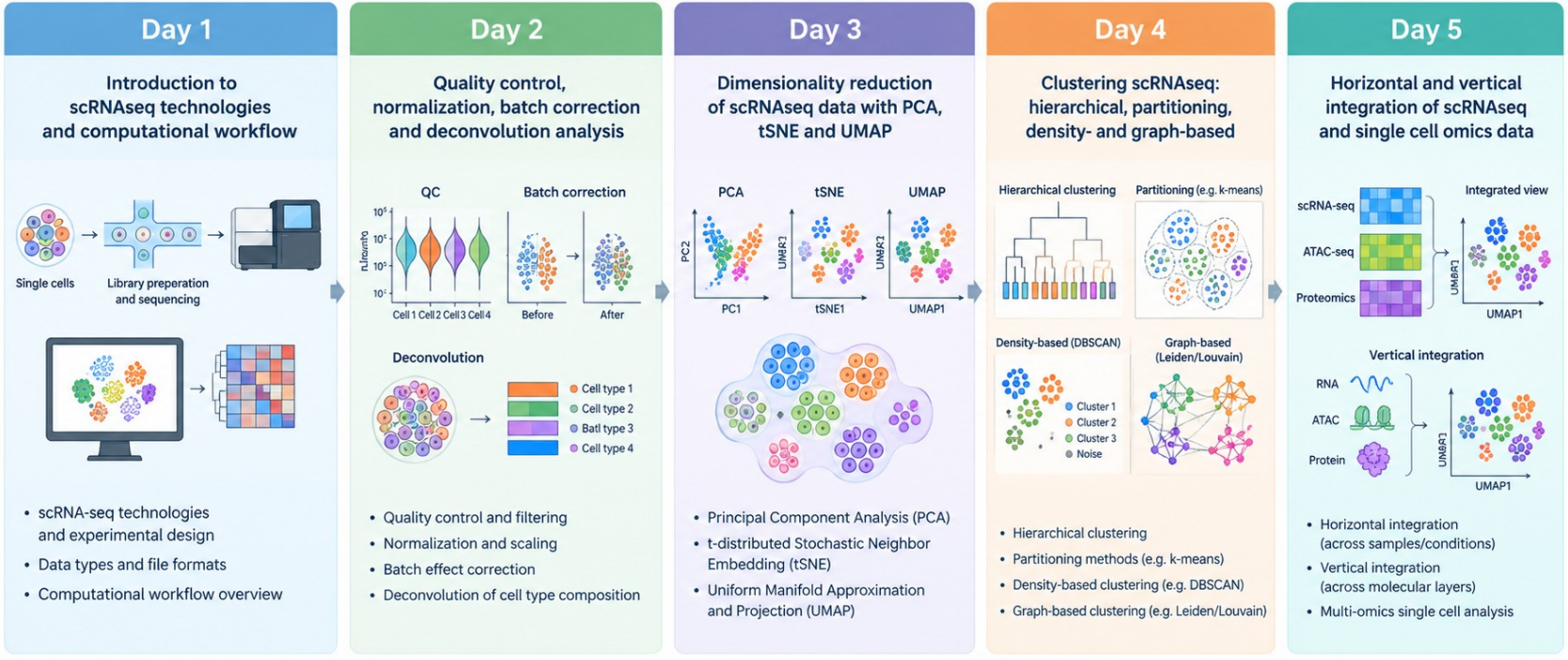


About you

- Name
- University / Institute / Company
- Research interest(s)
- Previous experience with computational analysis and bioinformatics
- Motivation to join the course
- Expectations from the course

Course outline

Day 1: introduction to scRNAseq technologies and computational Seurat workflow
Day 2: quality control, normalization, batch correction and deconvolution analysis
Day 3: dimensionality reduction of scRNAseq data with PCA, tSNE and UMAP
Day 4: clustering scRNAseq: hierarchical, partitioning, density- and graph-based
Day 5: horizontal and vertical integration of scRNAseq and single cell omics data



Housekeeping rules

- Please ask your questions anytime, it is ok to interrupt me, or you can raise your hand and ask. You can also ask questions online in the zoom chat or later by email.

- I kindly ask you to please keep your webcams on for better communication, this is of course not mandatory an up to you but will help me to adjust me explanations

- The course includes 10 lectures (~1 h each) and 10 practicals (~1.5 h each), there is a 15 min break after each lecture and 1 h lunch break every day of the course. We can potentially make longer breaks if needed.

- During practicals, we will use Rmarkdown and Jupyter notebooks. At the end of each practical, I will use rendered html-versions of the notebooks to go through command lines with my explanations.

- The material is based on data and problems from computational biology, however the concepts discussed are general and can be applied for other types of data.

Practical information: time and github repo

The course will take place from 2 pm to 8 pm UK time (3 pm to 9 pm CET)

The course GitHub repository containing lectures and exercises is:

https://github.com/NikolayOskolkov/SBW2022

Please bookmark this address!

SBW2022

Public

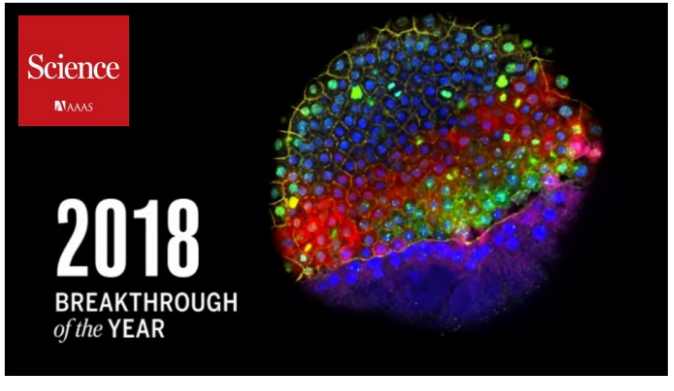
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NikolayOskolkov added articles cfe7839 · 33 minutes ago 6 Commits

articles	added articles	33 minutes ago
images	added logo	53 minutes ago
README.md	added articles	33 minutes ago
SBW2022_Lab_SkeletalMuscle_snRNAseq_Se...	Add files via upload	4 years ago
SBW2022_Lab_SkeletalMuscle_snRNAseq_Se...	Add files via upload	4 years ago
SkeletalMuscle_HumanChimp_10X.zip	Add files via upload	4 years ago
scRNAseq_data_analysis_Oskolkov.ppt	Add files via upload	4 years ago

README



Single Cell RNAseq Analysis

Instructor

About

This is a teaching material for scRNAseq workshop within SBW2022

Readme
Activity
2 stars
1 watching
1 fork

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 1

NikolayOskolkov Nikolay Oskolkov

Languages

HTML 100%

Suggested workflows

Based on your tech stack

SLSA Generic generator Configure
Generate SLSA3 provenance for your existing release workflows
By Open Source Security Foundation (OpenSSF)

Jekyll using Docker Configure

Google Colab

Python based



Python
(with rich ecosystem)

Google Drive
(easy file access)

Free GPU/TPU
(for ML and deep learning)

No installation
(runs in the browser)

VS

Post Cloud

RStudio and R based



R
(statistical computing)

RStudio
(full IDE)

Dedicated servers
(stable and secure)

Large storage
(for big datasets)

Key strengths

- Easy to get started (no setup)
- Great for Python and machine learning
- Free access to GPUs/TPUs
- Seamless integration with Google Drive
- Ideal for notebooks and reproducible workflows

Limitations

- Python only (no RStudio)
- Session time limits (can disconnect)
- Less control over the computing environment
- Not ideal for large-scale or long-running jobs

Key strengths

- RStudio with full IDE and packages
- Great for statistical analysis and visualization
- Stable, dedicated computing environment
- Suitable for large datasets and long-running analyses
- Supports collaborative and reproducible workflows

Limitations

- R only (no Python)
- No free GPU/TPU access (CPU only)
- Requires access to Post Cloud infrastructure
- Slightly more setup and learning curve for beginners

Both platforms support your research – choose the one that fits your needs

Google Colab for flexible, Python-based, cloud computing | Post Cloud for RStudio, R and dedicated resources

posit.cloud/content/13008494

Account owners can now enable **Posit Assistant**, an AI coding assistant for RStudio projects. Get started quickly with a free trial Posit AI subscription. **Learn more.**

Posit Cloud

Your Workspace / PopGen_PopStructure

Nikolay Oskolkov

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Untitled1*

```
gen <- t(matrix(c(1, 0, 2, 0, 2, 0, 2, 1, 1, 1, 0, 1, 0, 2, 1, 2, 1, 1, 1, 1, 1, 0, 1, 0, 2, 0, 1, 1, 0, 2
colnames(gen) <- paste0("Ind", 1:5)
rownames(gen) <- paste0("SNP", 1:7)
print(gen)

gen_centered <- scale(gen, center = TRUE, scale = FALSE) # mean centering
covariance <- t(gen_centered) %*% gen_centered # computing variance-covariance matrix
eig <- eigen(covariance) # performing eigen value decomposition of the variance-covariance matrix

barplot(eig$values / sum(eig$values), names = paste0("PC", 1:5))

```

11:1 (Top Level) R Script

Environment History Connections Tutorial

Import Dataset 143 MiB List

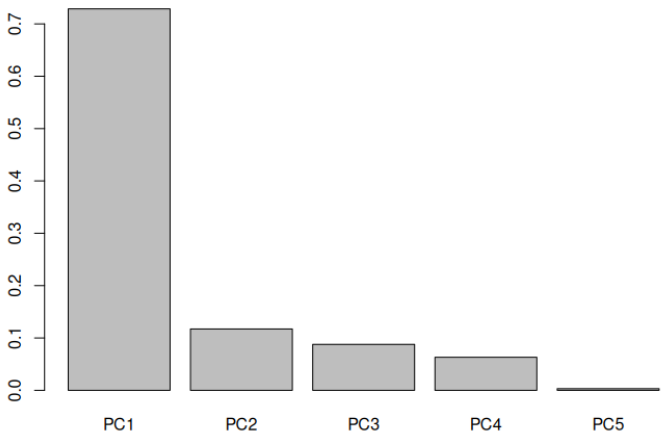
R Global Environment

Data

covariance	num [1:5, 1:5] 6 3 -1 -3 -4 ...
eig	List of 2
gen	num [1:7, 1:5] 1 0 2 0 2 0 2 1 1 1 ...
gen_centered	num [1:7, 1:5] 0 -1 1 -1 1 ...

Files Plots Packages Help Viewer Presentation

Zoom Export Publish



Console Terminal Background Jobs

R 4.6.1 · /cloud/project/

```
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Session restored from your saved work on 2026-Sep-27 14:27:56 UTC (3 hours ago)
> gen <- t(matrix(c(1, 0, 2, 0, 2, 0, 2, 1, 1, 1, 0, 1, 0, 2, 1, 2, 1, 1, 1, 1, 1, 0, 1, 0, 2, 0, 1, 1, 0, 2,
1, 2, 0, 1, 0), 5, by = TRUE))
> colnames(gen) <- paste0("Ind", 1:5)
> rownames(gen) <- paste0("SNP", 1:7)
> print(gen)
     Ind1 Ind2 Ind3 Ind4 Ind5
SNP1    1    1    1    0    0
SNP2    0    1    2    1    2
SNP3    2    1    1    0    1
SNP4    0    0    1    2    2
SNP5    2    1    1    0    0
SNP6    0    0    1    1    1
SNP7    2    2    1    1    0
> gen_centered <- scale(gen, center = TRUE, scale = FALSE) # mean centering
> covariance <- t(gen_centered) %*% gen_centered # computing variance-covariance matrix
> eig <- eigen(covariance) # performing eigen value decomposition of the variance-covariance matrix
> barplot(eig$values / sum(eig$values), names = paste0("PC", 1:5))
>
```

To run Jupyter notebooks on Google Colab please do the following:

1. Clone this repo: https://github.com/NikolayOskolkov/SBW2022
2. Upload the repo to your Google drive
3. Start the Jupyter notebook from the Google drive and mount your google drive:

```
#Mount Google Drive to Colab

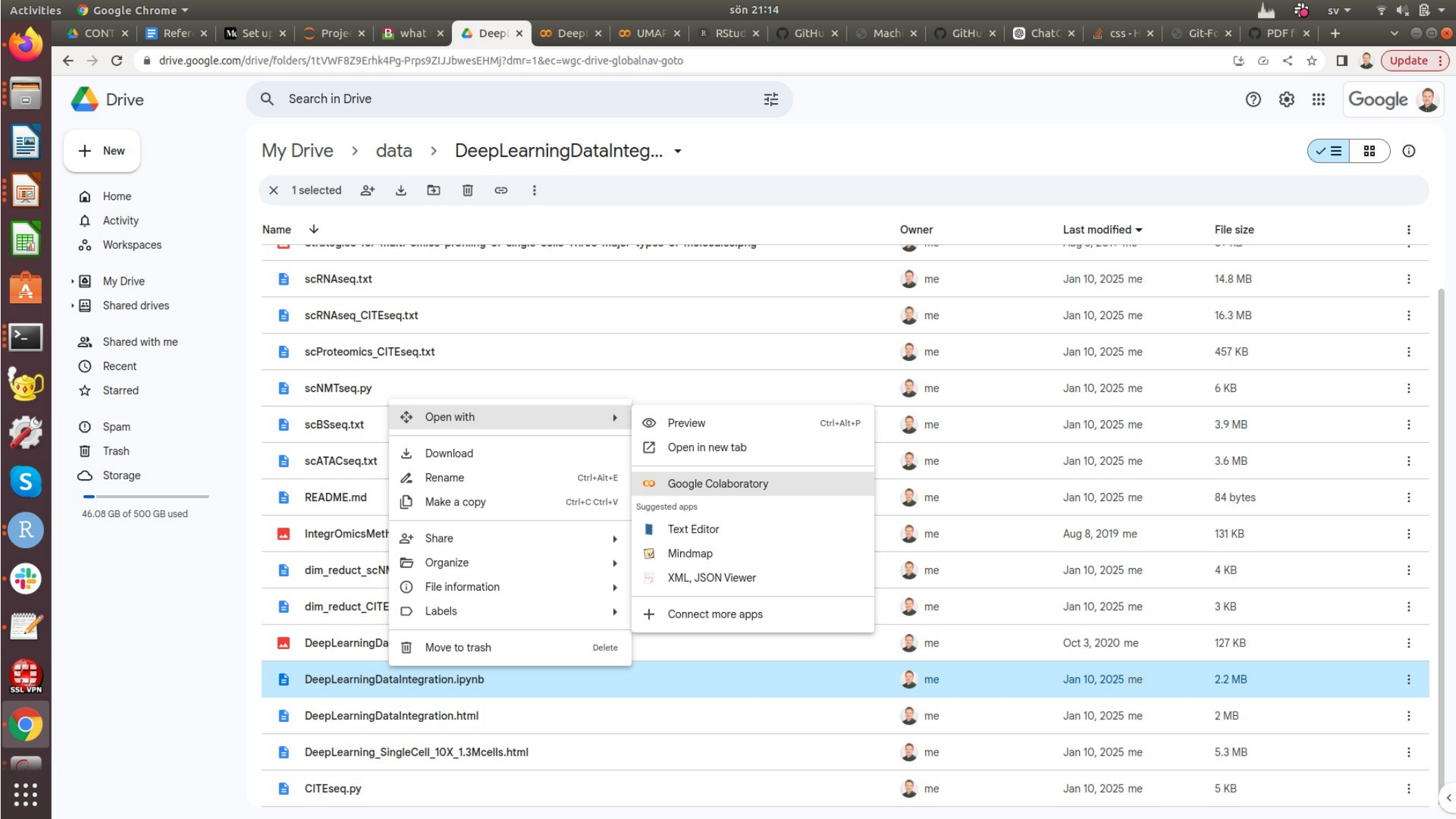
from google.colab import drive
drive.mount('/content/drive')

!ls -l /content/drive/MyDrive/

import os
os.chdir("/content/drive/MyDrive/data")
```

N.B. you need a Google account to be able to run Google Colab!




Activities
Google Chrome
sön 21:14
drive.google.com/drive/folders/1tVWF8Z9Erhk4Pg-Prps9ZIJJbwesEHMj?dmr=1&ec=wgc-drive-globalnav-goto
Update
Drive
Search in Drive
Google
New
Home
Activity
Workspaces
My Drive
Shared drives
Shared with me
Recent
Starred
Spam
Trash
Storage
46.08 GB of 500 GB used
My Drive › data › DeepLearningDataInteg...
1 selected
Name
Owner
Last modified
File size
scRNAseq.txt me Jan 10, 2025 me 14.8 MB
scRNAseq_CITEseq.txt me Jan 10, 2025 me 16.3 MB
scProteomics_CITEseq.txt me Jan 10, 2025 me 457 KB
scNMTseq.py me Jan 10, 2025 me 6 KB
scBSseq.txt me Jan 10, 2025 me 3.9 MB
scATACseq.txt me Jan 10, 2025 me 3.6 MB
README.md me Jan 10, 2025 me 84 bytes
IntegrOmicsMeth me Aug 8, 2019 me 131 KB
dim_reduct_scN me Jan 10, 2025 me 4 KB
dim_reduct_CITE me Jan 10, 2025 me 3 KB
DeepLearningDa me Oct 3, 2020 me 127 KB
DeepLearningDataIntegration.ipynb me Jan 10, 2025 me 2.2 MB
DeepLearningDataIntegration.html me Jan 10, 2025 me 2 MB
DeepLearning_SingleCell_10X_1.3Mcells.html me Jan 10, 2025 me 5.3 MB
CITEseq.py me Jan 10, 2025 me 5 KB
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XML, JSON Viewer
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colab.research.google.com/drive/1Q57pB5IT92sRKQEuGERQSgdMVHmC6qZd#scrollTo=HLDfYRzqMzAe Update

DeepLearningDataIntegration.ipynb

File Edit View Insert Runtime Tools Help All changes saved

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Deep Learning for Data Integration

Biological and biomedical research has been tremendously benefiting last decade from the technological progress delivering DNA sequence (genomics), gene expression (transcriptomics), protein abundance (proteomics) and many other levels of biological information commonly referred to as OMICs. Despite individual OMICs layers are capable of answering many important biological questions, their combination and consequent synergistic effects from their complementarity promise new insights into behavior of biological systems such as cells, tissues and organisms. Therefore OMICs integration represents the contemporary challenge in Biology and Biomedicine.

```
[ ] from google.colab import drive
    drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
[ ] !ls -l /content/drive/My\ Drive/data/DeepLearningDataIntegration
```

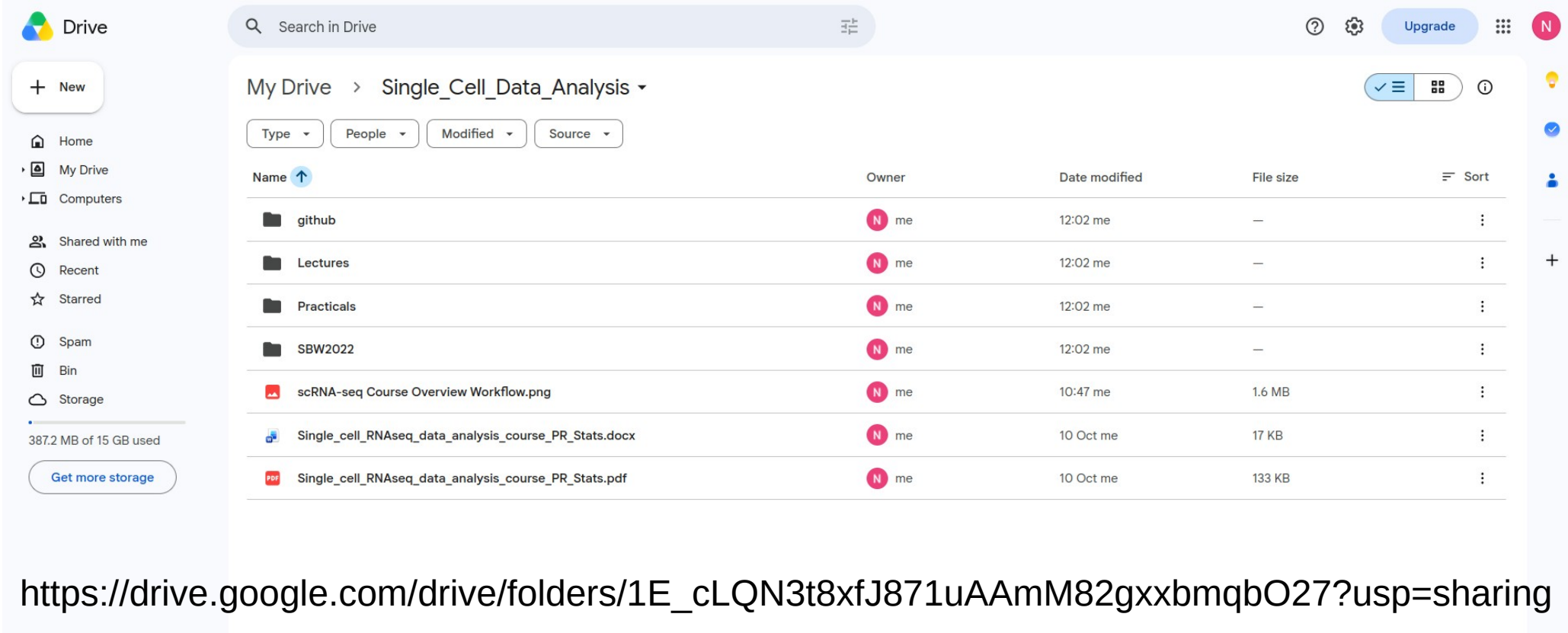
```
total 49588
-rw------- 1 root root     5105 Jan 10 12:26 CITEseq.py
-rw------- 1 root root  2066933 Jan 10 12:26 DeepLearningDataIntegration.html
-rw------- 1 root root  1747228 Jan 10 12:41 DeepLearningDataIntegration.ipynb
-rw------- 1 root root   130233 Oct  3  2020 DeepLearningDataIntegration.jpg
-rw------- 1 root root  5598515 Jan 10 12:26 DeepLearning_SingleCell_10X_1.3Mcells.html
-rw------- 1 root root     3499 Jan 10 12:26 dim_reduct_CITEseq.py
-rw------- 1 root root     3616 Jan 10 12:26 dim_reduct_scNMTseq.py
-rw------- 1 root root   134229 Aug  8  2019 IntegrOmicsMethods.png
-rw------- 1 root root       84 Jan 10 12:26 README.md
-rw------- 1 root root  3772209 Jan 10 12:26 scATACseq.txt
-rw------- 1 root root  4082712 Jan 10 12:26 scBSseq.txt
-rw------- 1 root root     6141 Jan 10 12:26 scNMTseq.py
-rw------- 1 root root   467570 Jan 10 12:26 scProteomics_CITEseq.txt
-rw------- 1 root root 17128751 Jan 10 12:26 scRNAseq_CITEseq.txt
-rw------- 1 root root 15547715 Jan 10 12:26 scRNAseq.txt
-rw------- 1 root root    71008 Aug  8  2019 Strategies-for-multi-omics-profiling-of-single-cells-Three-major-types-of-molecules.png
-rw------- 1 root root     3647 Jan 10 12:26 tSNE_on_Autoencoder_CITEseq.py
-rw------- 1 root root     3635 Jan 10 12:26 tsne_on_autoencoder_scNMTseq.py
```

```
import os
os.chdir("/content/drive/My Drive/data/DeepLearningDataIntegration")
```

```
[ ] from IPython.display import Image
    Image('DeepLearningDataIntegration.jpg', width=2000)
```

Connected to Python 3 Google Compute Engine backend

Course material is available via Google Drive



https://drive.google.com/drive/folders/1E_cLQN3t8xfJ871uAAmM82gxxbmqbO27?usp=sharing

Please let me know at nikolay.oskolkov@niri.lv if you can't access it!